CENTER FOR DRUG EVALUATION AND RESEARCH

APPLICATION NUMBER:

209184Orig1s000

<u>CLINICAL PHARMACOLOGY AND BIOPHARMACEUTICS REVIEW(S)</u>

Office of Clinical Pharmacology Review

NDA Number	209184
Link to EDR	\\cdsesub1\evsprod\nda209184
Submission Date	12/5/2017
Submission Type	505(b)(2)
Brand Name	INBRIJA™
Generic Name	Levodopa Inhalation Powder (CVT-301)
Dosage Form and Strength	INBRIJA dry powder capsules contain 42 mg levodopa for use with the INBRIJA inhaler
Route of Administration	Oral Inhalation
Proposed Dose:	Oral inhalation of the contents of two 42 mg capsules (84 mg) as needed, up to 5 times a day.
Proposed Indication	Treatment of symptoms of OFF periods in Parkinson's disease as an adjunct to a carbidopa-levodopa regimen
Applicant	Acorda Therapeutics, Inc.
Associated IND	115750
OCP Review Team	Mariam Ahmed, PhD; Kevin Krudys, PhD; Sreedharan Sabarinath, PhD

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1. EXECUTIVE SUMMARY

The applicant, Acorda Therapeutics, Inc., submitted a New Drug Application (NDA) for CVT-301 (INBRIJA™, levodopa for oral inhalation) through the 505(b)(2) regulatory pathway. The proposed indication is treatment of symptoms of OFF periods in Parkinson's disease (PD) as an adjunct to a daily oral carbidopa and levodopa (LD) regimen. The applicant is relying on literature reports and FDA's findings on safety of the approved prescription product, SINEMET® oral tablets (NDA 017555) as the reference listed drug (RLD). SINEMET® is approved for the treatment of PD, post-encephalitic Parkinsonism, and symptomatic Parkinsonism that may follow carbon monoxide intoxication or manganese intoxication.

Acorda conducted 11 clinical studies with CVT-301 to support the new formulation, delivery system, route of administration, and the proposed indication. These included three Phase 1 studies in healthy volunteers (CVT-301-001, CVT-301-006, and CVT-301-010), two Phase 1 studies in special populations (CVT-301-007 in smokers and CVT-301-008 in asthmatics), two Phase 2 studies in PD patients (CVT-301-002 and CVT-301-003), and three Phase 3 studies (pivotal study: CVT-301-004, and 2 ongoing long-term safety studies: CVT-301-004E, and CVT-301-005). All Phase 3 studies used the to-be-marketed product.

The pivotal Phase 3 study (CVT-301-004) was double-blinded, placebo-controlled that evaluated two dose levels: 60 and 84 mg of levodopa oral inhalation. The study met its primary endpoint of change from pre-dose Unified Parkinson's Disease Rating Scale Part 3 (UPDRS Part III) score at 30 minutes for the 84 mg dose group as compared to the placebo group at Week 12.

Study CVT-301-010 was a Phase 1, single dose pharmacokinetic (PK) bridging study that compared 60 and 84 mg strengths of CVT-301 with the RLD SINEMET® (25-100 mg) tablets.

The primary objectives of this review are:

- To evaluate the PK bridging of the proposed product to the RLD (SINEMET® oral tablets) and,
- To assess the adequacy of the proposed dosing recommendations for general and special populations

1.1 Recommendations

The Office of Clinical Pharmacology has reviewed the NDA and the application is acceptable from OCP perspective.

Review Issue	Recommendations and Comments
Pivotal or supportive evidence of effectiveness	Primary evidence of effectiveness was established in a pivotal Phase 3 efficacy study, CVT-301-004, based on change in UPDRS Part III score from pre-dose to 30 minutes post-dose at Week 12. The Phase 2b study CVT-301-003 in PD patients provides supportive evidence of effectiveness.
General dosing instructions	The proposed dose of INBRIJA™ is 84 mg (as two 42 mg capsules) for oral inhalation, as needed, for the treatment of symptoms of OFF periods, up to 5 times a day. Clinical experience with patients taking more than 3 doses per day is limited. The maximum daily dose of INBRIJA™ should not exceed 420 mg. Only one inhalation dose (as 2 capsules) is allowed per OFF period.
Dosing in patient subgroups (intrinsic and extrinsic factors)	The effects of renal and hepatic impairments were not studied with INBRIJA™, and no specific dosing instructions are provided for these populations in the RLD label. No dose adjustment is required for smokers.
Labeling	The proposed labeling in general is acceptable.
Bridge between the to-be-marketed and clinical trial formulations	The to-be-marketed formulation is the same as the formulation used in the Phase 3 efficacy and safety trials and in the PK bridging study (CVT-301-010).

1.2 Post-Marketing Requirements and Commitments

None.

2. SUMMARY OF CLINICAL PHARMACOLOGY ASSESSMENT

2.1 Pharmacology and Clinical Pharmacokinetics

Levodopa, the metabolic precursor of dopamine, crosses the blood-brain barrier, and is presumably converted to dopamine in the brain. This is thought to be the mechanism whereby levodopa relieves symptoms of Parkinson's disease[1].

After administration of INBRIJA™ 84 mg (oral inhalation of two 42 mg capsules), the median time to maximum plasma concentration (T_{max}) was achieved at 30 min. Apparent volume of distribution (Vz/F) was 168 L and the apparent terminal elimination half-life ($t_{1/2}$) was 2.3 hours. Levodopa is extensively metabolized. The two major metabolic pathways are decarboxylation by dopa decarboxylase and O-methylation by catechol-O-methyltransferase (COMT) enzymes.

[1] SINEMET® USPI: https://www.accessdata.fda.gov/drugsatfda_docs/label/2014/017555s056s068s071lbl.pdf

The increase in plasma LD exposure was proportional to the INBRIJA™ dose over the two studied doses of 60 and 84 mg. The bioavailability of INBRIJA™ 84 mg oral inhalation relative to SINEMET® (carbidopa-LD) 25-100 mg oral tablets is ~70%.

2.2 Dosing and Therapeutic Individualization

2.2.1 General dosing

The proposed dose of INBRIJATM is 84 mg (as two 42 mg capsules) for oral inhalation, as needed, for the treatment of symptoms of OFF periods, up to 5 times a day. However, there is limited clinical experience with patients taking more than 3 doses per day. The maximum daily dose of INBRIJA™ should not exceed 420 mg. Only one inhalation dose (as 2 capsules) is allowed per OFF period.

2.2.2 Therapeutic individualization

No dose adjustment is necessary in smokers (≥10 cigarettes/day for at least 12 months). From a PK perspective, there is no difference in INBRIJA™ exposure in patients with asthma as compared to healthy volunteers. However, because of the risk of bronchospasm, use of INBRIJA™ in patients with asthma is not recommended.

2.3 Outstanding Issues

None.

2.4 Summary of Labeling Recommendations

The Office of Clinical Pharmacology has reviewed the package insert of INBRIJA™ and finds it generally acceptable.

3. COMPREHENSIVE CLINICAL PHARMACOLOGY REVIEW

3.1 Overview of the Product and Regulatory Background

CVT-301 is a drug-device combination that consists of: (1) CVT-301 capsules containing a dry powder form of levodopa and (2) the CVT-301 dry powder inhaler. CVT-301 does not contain carbidopa.

The proposed indication is treatment of symptoms of OFF periods in Parkinson's disease as an adjunct to a carbidopa-levodopa regimen. Currently, APOKYN® subcutaneous injection is the only approved therapeutic option for this indication.

During pre-Investigational New Drug (IND) meeting, it was agreed that the filing of NDA for CVT-301 under Section505(b)(2) of the Food, Drug and Cosmetic Act based on the Agency's previous findings of nonclinical and clinical safety data on levodopa would be an acceptable approach. During the End-of-Phase 2 meeting, the agency agreed with the scientific rationale to establish a "bridge" based on PK comparison of CVT-301 to SINEMET® as the reference product. In addition, in that meeting a general agreement was reached on the Phase 3 clinical development program and design elements of the pivotal Phase 3 Study CVT-301-004. FDA also agreed that results from Phase 2b Study CVT-301-003 could be supportive for this NDA.

3.2 General Pharmacology and Pharmacokinetic Characteristics

The peak plasma concentrations of levodopa were achieved at about 30 min after oral inhalation of 84 mg dose of CVT-301. The apparent terminal elimination half-life ($t_{1/2}$) of levodopa was 2.3 hours following a single administration of 84 mg in the presence of carbidopa. All PK studies with CVT-301 used pretreatment doses of 50 mg of carbidopa every 8 hours prior to inhalation of CVT-301, because CVT-301 is an adjunct therapy in patients already on carbidopa-LD regimen. The carbidopa dosing regimen for the PK studies was selected to ensure that maximal inhibition of aromatic L-amino acid decarboxylase was achieved. It has been reported that a minimum of 75-150 mg per day of carbidopa is required to achieve maximal decarboxylase inhibition[2].

The bioavailability of INBRIJA™ 84 mg relative to SINEMET® (carbidopa-LD) 25-100 mg oral tablets is ~70%. Dose proportionality for the to-be-marketed product was established between the CVT-301 60 mg and 84 mg doses. For more details, please refer to Section 4.2.

3.3 Clinical Pharmacology Review Questions

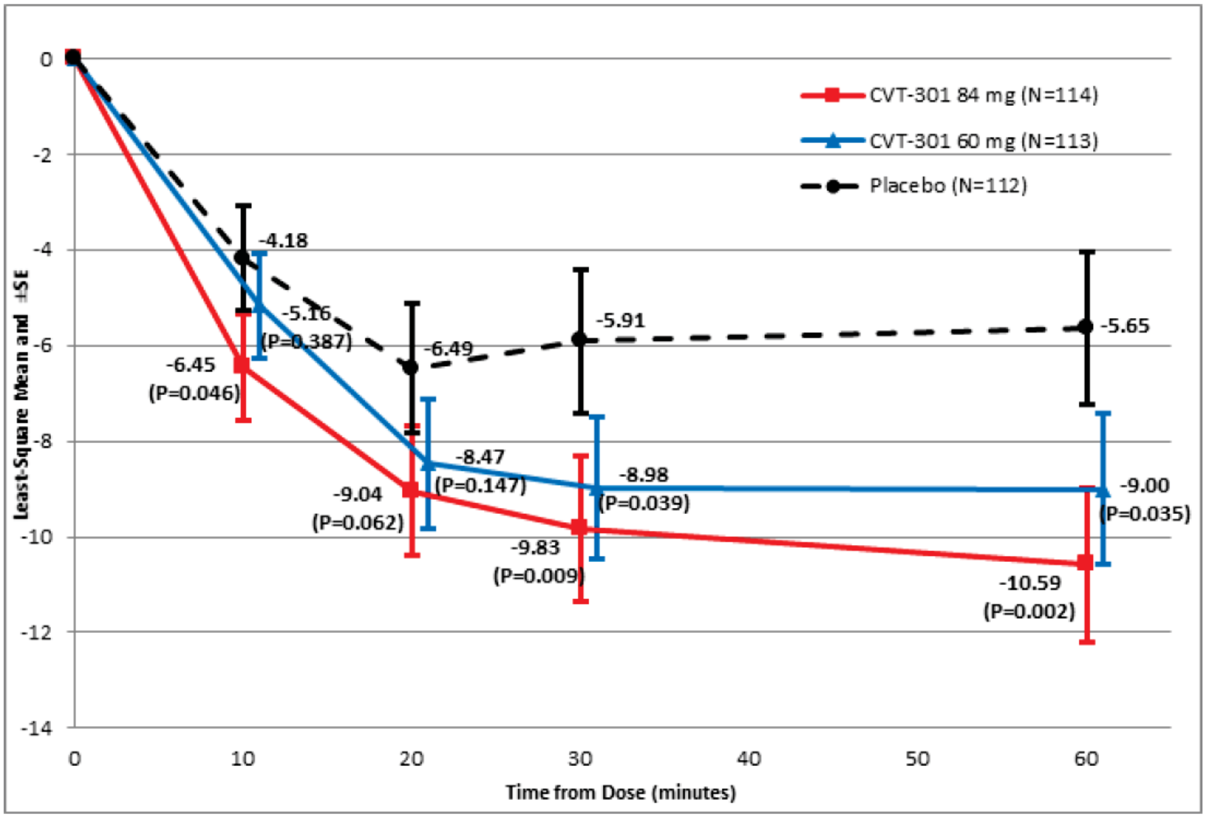
3.3.1 To what extent does the available clinical pharmacology information provide pivotal or supportive evidence of effectiveness?

Evidence of effectiveness was established from the pivotal efficacy study CVT-301-004. A Phase 2b study CVT-301-003 in PD patients was considered as supportive.

Study CVT-301-004 was a Phase 3 randomized, double-blind, placebo-controlled, 12-week, multicenter (North America and Europe) study that assessed the efficacy of inhaled CVT-301 for the treatment of up to five OFF periods per day in PD subjects experiencing motor fluctuations (OFF periods) (N=339). PD patients receiving carbidopa-LD for management of their symptoms were randomly assigned in a 1:1:1 ratio to receive inhaled CVT-301 60 mg, CVT-301 84 mg, or placebo. Patients were to remain on their background PD treatment regimen. The primary endpoint was the change from pre-dose in UPDRS Part III score at 30 minutes following the 84-mg dosed at OFF period at Week 12. The study met its primary endpoint. In addition, the applicant reported numerical improvement in the UPDRS Part III score from pre-dose as early as 10 minutes following the 84-mg dose with nominal P-value <0.05 (Figure 1).

[2] Arch Neurol. 1980 Mar;37(3):146-9.

Figure 1: Change in the UPDRS Part III Score from Pre-dose to 10, 20, 30, and 60 Minutes Post-dose at Week 12 for Both CVT-301 Doses and Placebo (Study CVT-301-004, ITT Population)



Note: P-value on the figure indicates nominal P-value.
Source: 2.7.3 Summary of Clinical Efficacy, Figure 7.

The supportive study CVT-301-003 was a Phase 2b, randomized, double-blind, placebo-controlled, multicenter (North America and Europe) study that assessed the efficacy of inhaled CVT-301 for the treatment of up to three OFF periods per day in PD subjects experiencing motor fluctuations. Subjects (N=86) were randomly assigned in a 1:1 ratio to receive either placebo or CVT-301 55.2 mg for 2 weeks followed by increased CVT-301 dose to 82.8 mg for another 2 weeks. The primary endpoint was the change from pre-dose in the average UPDRS Part III score at 10-60 minutes (i.e. 10, 20, 30, and 60 minutes) post-dose at Week 4. As per applicant's analysis, there was a statistically significant reduction in the average UPDRS Part III score 10 to 60 minutes post-dose compared with placebo (least square mean (LSM) difference = -6.95; p < 0.001). For more details, please refer to Dr. Susanne Goldstein's clinical review and Dr. Sungwon Lee's statistical review.

3.3.2 Is the proposed dosing regimen appropriate for the general patient population for which the indication is being sought?

The applicant proposes 84-mg dose for the treatment of up to 5 OFF periods per day. However, there is limited clinical experience with patients taking more than 3 doses per day. In the pivotal Phase 3 efficacy study, 2 dose levels of CVT-301 (60 and 84 mg/OFF period for treatment of up to 5 OFF periods per day) were compared to placebo at Week 12. Figure 1 above shows the change from pre-dose in UPDRS Part III score at Week 12 for the CVT-301 84 mg, CVT-301 60 mg, and placebo groups over 1 hour post-dose. These results suggest that proposed dose of 84 mg is effective relative to placebo and the effect on UPDRS Part III score reduction is observed from 10 minutes onwards. The 60-mg dose resulted in reduction in UPDRS Part III score that was numerically lower as compared to the 84-mg dose at every post-dose time point (LSM difference~1).

Per applicant analyses, the safety findings from the pivotal Phase 3 study and the ongoing long-term safety studies (CVT-301-004E and CVT-301-005) appeared comparable across the 2 dose levels. Overall, 43.8%, 56.6%, and 57.9% patients in the placebo, CVT-301 60 mg, and CVT-301 84 mg groups, respectively, experienced at least 1 treatment emergent adverse event (TEAE) during the study. Reported TEAEs were generally consistent with known AEs listed for the reference listed drug (SINEMET®), with the addition of pulmonary-related AEs. The most frequently reported TEAE was cough (40 TEAEs in 36 patients), which was observed at a lower frequency in the placebo group (1.8%) compared with the CVT-301 60 mg (15.0%) and CVT-301 84 mg (14.9%) groups. Please refer to the clinical review for additional details.

3.3.3 Is an alternative dosing regimen and/or management strategy required for subpopulations based on intrinsic/extrinsic factors?

The effect of gender, asthma, and smoking on levodopa exposures after oral inhalation of INBRIJA™ were assessed in this NDA. Alternative dosing regimen and management strategy is not required based on gender or smoking status (smokers were defined as smoking≥10 cigarettes/day for at least 12 months). For more details, please refer to Sections 4.3, and 4.4. From a PK perspective, there is no difference in INBRIJA™ exposure in patients with asthma as compared to healthy volunteers. However, because of the risk of bronchospasm, use of INBRIJA™ in patients with asthma is not recommended.

The effect of food on absorption was not studied. While the amount swallowed after CVT-301 oral inhalation was studied in this application, literature reports indicate that there is a potential for swallowing up to 40-90% of the oral inhaled drug products[3]. In the pivotal and supportive efficacy trials, INBRIJA™ was administered without regard to food. Therefore, INBRIJA™ can be administered without regard to food.

The effect of other intrinsic and extrinsic factors, such as renal and hepatic impairment, and drug interactions, were not specifically studied in this application. However, they are expected to be the same as with the RLD product. Please refer to SINEMET® USPI for information on drug-drug interactions with levodopa.

[3]AAPS J. 2015 May; 17(3): 769–775.

3.3.4 Is the PK bridging provided in the application adequate?

Bridging between the clinical trial formulations and the to-be-marketed formulation:

The to-be-marketed CVT-301 capsule was used in the pivotal Phase 3 study (CVT-301-004) and was produced through a scale-up of the formulation used in the supportive Phase 2b study (CVT-301-003). The composition of the capsules remained unchanged across all clinical trials [4]. Moreover, except for optimization of some inhaler external design features, the CVT-301 dry powder inhaler also remained unchanged between the supportive Phase 2b and the pivotal efficacy and safety Phase 3 studies.

No PK data was collected during the supportive Phase 2b or the pivotal Phase 3 studies. However, the to-be-marketed formulation's PK was characterized in healthy subjects in a Phase 1 CVT-301-010 PK bridging study and provides adequate PK bridging to the RLD.

According to the biopharmaceutics and CMC review teams, the scale-up process of the Phase 2b capsule is not expected to significantly affect the PK of levodopa and the slight changes in the total weight between Phase 2b and Phase 3 capsules is not expected to affect the bioavailable dose. The fine particle dose (FPD) for these formulations in the Phase 2b and Phase 3 studies are reported to be the same. For more details, please refer to the biopharmaceutics and CMC reviews by Dr. Akm Khairuzzaman, and Dr. Dan Berger.

Bridging between the to-be-marketed formulation and the RLD SINEMET® (25-100 mg):

The applicant conducted Study CVT-301-010 to establish the PK bridge between the to-be-marketed CVT-301 formulation and the RLD. This was a Phase 1, single dose PK study that compared 60 and 84 mg strengths of CVT-301 with the RLD SINEMET® (25-100 mg) tablets. Based on this study, the systemic exposure (C_{max} and $AUC_{0\text{-}24h}$) of levodopa was lower following both 60 mg and 84 mg inhaled doses of CVT-301 than following an oral dose of SINEMET® 25-100 oral tablet. The ratios of baseline-unadjusted, dose-normalized- $AUC_{0\text{-}24h}$ and dose-normalized C_{max} following a CVT-301 84 mg dose were approximately 70% and 50%, respectively, of those following SINEMET® 25-100 dose. For more details, please refer to Section 4.2.

[4] CVT-301 capsule contains dry powder LD intended for oral inhalation using CVT-301 dry powder inhaler. The capsules composed of (b) (4) %w/w of LD, 1,2-dipalmitoyl-sn-glycero-3-phosphatidylcholine (DPPC) and sodium chloride (NaCL), filled into size 00 hydroxypropyl methylcellulose (HPMC) capsules.

4. APPENDICES

4.1 Summary of Bioanalytical Method Validation and Performance

PK data were collected in three Phase 1 studies in healthy volunteers (Studies CVT-301-001, CVT-301-006, and CVT-301-010) and in two Phase 1 studies in special populations (Study CVT-301-007 in smokers and non-smokers, and Study CVT-301-008, in asthmatics). PK data was also collected in Study CVT-301-002, a Phase 2a study that utilized a different than the to-be-marketed CVT-301 inhaler. The sponsor has submitted assay validation reports for these studies. In this review, we focused on the LD assay validation for studies CVT-301-007, CVT-301-008 and CVT-301-010.These studies were done with the same to-be-marketed product and are considered important to support labeling information.

Studies CVT-301-007, CVT-301-008 and CVT-301-010 evaluated the PK in smokers, asthma, and the relative bioavailability of the to-be-marketed formulation of CVT-301 relative to the RLD (SINEMET® 25-100 mg), respectively. For these studies, human plasma samples were analyzed for LD using a method by (b) (4) (415/25716HV). The range of the assay was 10-5000 ng/mL. The internal standard (IS) included in the assay was LD-d3. Summaries of the performance characteristics and validation attributes for plasma LD are provided in Table 1 below.

Table 1: Validation Study Summary for Method 415/25191HV Measuring Levodopa

Items	Results
Methodology	LC-MS/MS (isocratic)
Method Validation Report (MVR) Number	RD415/25716HV
Biological Matrix	Human plasma (K2, EDTA, also treated with carbidopa at 5µg/mL of plasma to inhibit decarboxylase activity and prevent enzymatic degradation of levodopa. Sodium metabisulfite (8% w/v): plasma at 1:10 v/v was added to stabilize levodopa
Analyte of Interest	Levodopa (LD)
Internal Standard (IS)	Levodopa (LD-d3)
Calibration Curve Range	10- 5000 ng/mL
Lower limit of quantitation	Accuracy 100.1%, Precision 5.3%
QC concentrations (ng/mL)	10, 30, 400, 2000, 4000 and 20000 ng/mL
Accuracy, Precision and Recovery	
Inter Batch (Between-run) accuracy	96.3 -102.7%
Inter Batch (Between-run) precision	2.5 - 12.1%
Intra Batch (Within-run) accuracy	97.2 - 111.8%

Intra Batch (Within-run) precision	1.8 - 5.8%
Recovery	Mean recovery 66.2%
Selectivity and Specificity	Small peaks present (endogenous) in all 12 samples with 1<20% of LLOQ
Sensitivity (6 different blank plasma samples fortified at LLOQ)	Accuracy 94.3%, precision 10.2%
Stability	
Long term stability of the stock solution and working solution	100 mg/100 mL (1000 ng/µL) in 1M hydrochloric acid: • Room Temperature (RT) for 6 hrs: -3.2%, • 24 days at -20°C:-0.7% 0.5 ng/µL in 1M hydrochloric acid: • RT for 4 hrs: -1.2%, • 45 days at -20°C -1.9%
Short term stability in biological matrix at room temperature, accuracy	18 hrs: QC2: +1.4%; QC5: +1.5% 6 hrs: QC2: +0.2%; QC5: +1.0%
Long term stability in biological matrix, accuracy	Up to 141 days at -70°C: QC2: +1.3%; QC5: -2.3%
Freeze and thaw stability, accuracy	4 cycles at -70°C: QC2: -7.6 to +7.5; QC5: -1.2% to +6.3

Source: Report RD415/25716HV.

Reviewer Comment:

The methods satisfied the criteria for method validation and application to routine analysis as per the Guidance for Industry: Bioanalytical Method Validation[5], and hence are acceptable.

4.2 Bridging to the RLD and Dose Proportionality of the To-be-marketed Product

Study Report: CVT-301-010

Title of the Study: A Phase I Single Dose Pharmacokinetic Bridging Study to Compare Two Dose Strengths of CVT-301 (Levodopa Inhalation Powder) With an Oral Dose of SINEMET® (Carbidopa-Levodopa) Tablets.

[5] Available from: [https://www.fda.gov/downloads/drugs/guidances/ucm070107.Pdf].

Objectives:

The primary objective of the study was to determine the fasted state relative bioavailability of 2 capsules of CVT-301 30 mg (total dose 60 mg) and 2 capsules of CVT-301 42 mg (total dose 84 mg) compared with SINEMET® (i.e. the RLD). The secondary objective was to assess the dose proportionality of CVT-301 after oral inhalation of 2 capsules of CVT-301 30 mg and 2 capsules of CVT-301 42 mg in fasted state.

Study Design:

This s was an open-label, randomized, 3-period, crossover study in healthy volunteers. The CVT-301 formulation used in this study is same as the to-be-marketed product and was also used in the pivotal Phase 3 study CVT-301-004 and in the safety studies CVT-301-004E and CVT-301-005.

All subjects received one dose of each of the following:

A. SINEMET® (carbidopa-LD) 25-100 mg tablet for oral administration

B. CVT-301 60 mg (2x 30 mg capsules) for oral inhalation

C. CVT-301 84 mg (2x 42 mg capsules) for oral inhalation

All subjects received carbidopa 50 mg 3-times daily for 6 days beginning on Day -1. On Day 1, 3, and 5, subjects were administered the assigned product 1 hour following the morning dose of carbidopa 50 mg (pre-dose carbidopa was reduced to 25 mg when the RLD was administered). Day 2, 4, and 6, were levodopa washout days.

Population:

Healthy volunteers (N=24) with approximately equal numbers of men and women were randomized and completed the study.

PK and Blood Sampling:

Blood samples (a total of 15 per subject) were collected over 24 hours for the analysis of plasma concentrations of levodopa in each dosing period.

Statistical Methods:

The PK parameters were calculated using both baseline-unadjusted and baseline-adjusted concentration values. Since levodopa is an endogenous molecule, for baseline-adjusted PK parameters, the absolute pre-dose concentration was subtracted from post-dose levodopa concentrations.

Relative bioavailability was assessed by a repeated measures Analysis of Variance (ANOVA) model, with treatment, treatment sequence, and period as fixed effects, and subject nested within sequence as a random effect. This model was used to analyze the dose-normalized (DN) and unadjusted C_{max} and AUC_{0-24h}. Two-sided 90% confidence intervals (CIs) were calculated for the ratios between the investigational products (CVT-301 60 mg and 84 mg levodopa) and the RLD.

Results:

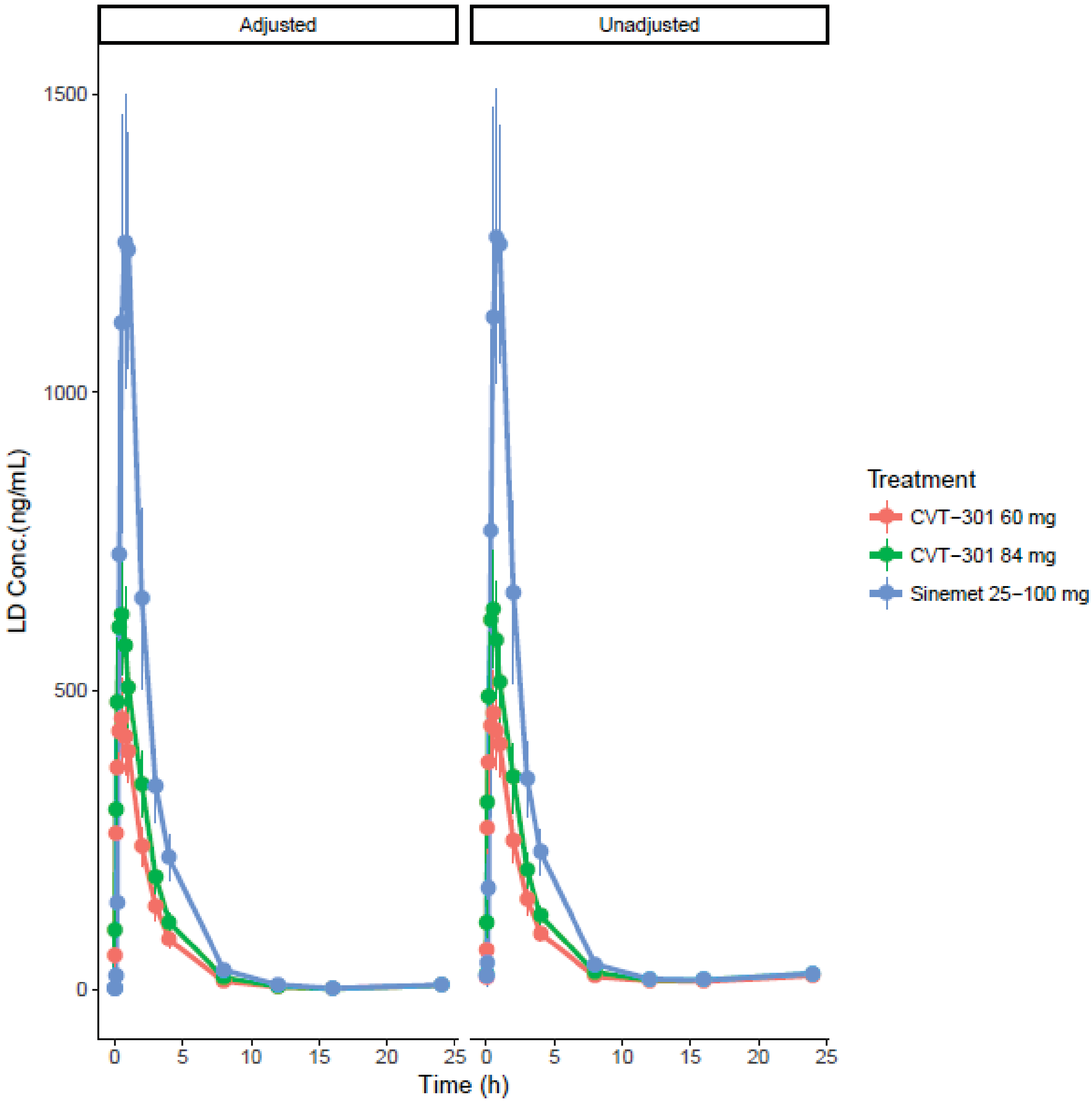
Baseline-adjusted PK parameters of LD are presented in Table 2. The mean and 95% confidence interval of LD concentration over 24 hours following administration of CVT-301 and SINEMET® 25-100 oral dose is presented in Figure 2.

Table 2: Summary of Baseline-adjusted Pharmacokinetic Parameters for LD

PK Parameter (Unit)	Treatment		
	SINEMET® 25-100 mg (N=24)	**CVT-301 60 mg (N=24)**	**CVT-301 84 mg (N=24)**
AUC_{0-4h} (h•ng/mL)	2390 (28.9)	965 (23.6)	1280 (34.3)
DN-AUC_{0-4h} (h•ng/mL/mg)	23.9 (28.9)	16.1 (23.6)	15.2 (34.3)
AUC_{0-24h} (h•ng/mL)	2880 (29.2)	1170 (23.2)[b]	1630 (30.1)[b]
DN-AUC_{0-24h} (h•ng/mL/mg)	28.8 (29.2)	19.6 (23.2)[b]	19.4 (30.1)[b]
C_{max} (ng/mL)	1630 (35.9)	522 (28.9)	696 (35.1)
DN-C_{max} (ng/mL/mg)	16.3 (35.9)	8.70 (28.9)	8.28 (35.1)
C_{10min} (ng/mL)	64.6 (333.9)	342 (42.7)	418 (59.2)
C_{min} (ng/mL)	9.61 (196.8)[b]	9.86 (95.5)	8.98 (126.4)
T_{max}(minute)[a]	45 (19-122)	30 (10-120)	30 (10-120)
$t_{1/2}$(h)[b]	1.88 (49.6)	1.96 (75.7)	2.30 (40.4)
V/F (L)[b]	96.2 (54.4)	144 (73.2)	168 (49.2)

Note: Geometric mean (CV%) data are presented unless otherwise noted.
Abbreviations: AUC = area under plasma concentration-time curve; C_{10min} = plasma concentration at 10 minutes post-dose; C_{max} = maximum plasma concentration; C_{min} = minimum plasma concentration; CV = coefficient of variation; DN = dose normalized; N = number of subjects; $t_{1/2}$ = apparent terminal elimination half-life; T_{max} = time to maximum plasma concentration; V/F = apparent terminal volume of distribution.
[a] Median (range).
[b] N < 24
Source: Study CVT-301-010 CSR, Table 7.

Figure 2: Baseline-adjusted and Baseline-unadjusted Plasma Concentration of LD versus Time (Mean and 95%Confidence Interval)



Source: Figure created by the reviewer

Statistical analyses of baseline-unadjusted dose normalized exposures (DN-$AUC_{0\text{-}24h}$ and DN-C_{max}) to assess the relative bioavailability of levodopa following oral doses of SINEMET® 25-100 and inhaled doses of CVT-301 60 mg and CVT-301 84 mg levodopa are presented in Table 3.

Table 3: Statistical Analysis of Dose-Normalized AUC_{0-24h} and C_{max} for Levodopa following Oral Doses of SINEMET® 25-100 and Inhaled Doses of 60 mg and 84 mg CVT-301- Unadjusted for Baseline

PK Parameter (Unit)	Treatment	Treatment vs. SINEMET® 25-100 mg	
		Ratio (%)[a]	90% CI
DN-C_{max} (ng/mL/mg)	CVT-301 60 mg	53.9	47.1-61.8
	CVT-301 84 mg	51.3	44.8-58.7
DN-AUC_{0-24h} (h•ng/mL/mg)	CVT-30160 mg	70.8	64.6-77.6
	CVT-301 84 mg	69.1	63.1-75.7

[a] ratio of dose normalized AUC represents the bioavailability of CVT-301 60 mg and CVT-301 84 mg relative to SINEMET® 25-100 mg.
Source: CVT-301-010 CSR, Table 9.

Statistical analyses of the dose proportionality of baseline-unadjusted values of dose-normalized-AUC_{0-24h} and dose-normalized C_{max} for LD following inhaled doses of CVT-301 60 mg and 84 mg are presented in Table 4.

Table 4: Dose Proportionality of Inhaled Doses of CVT-301 60 mg and 84 mg LD Assessed by Dose-Normalized AUC_{0-24h} and C_{max} for LD – Unadjusted for Baseline

PK Parameter (Unit)	CVT-301 84 mg vs. CVT-301 60 mg	
	Ratio (%)	90% CI for the Ratio
DN-C_{max} (ng/mL/mg)	95.1	85.6 - 105.8
DN-AUC_{0-24h}	97.2	88.5 - 106.8

Source: Study CVT-301-010 CSR, Table 10.

Reviewer Comments:

- *The overall study design is acceptable. The analytical assay method and sample analyses are also acceptable.*
- *Systemic exposure (C_{max} and AUC_{0-24h}) of levodopa was lower following both 60 mg and 84 mg inhaled doses of CVT-301 than following an oral dose of SINEMET® 25-100 oral tablet.*
- *The applicant's primary analysis focused on the baseline-unadjusted data. Usually, one would consider the baseline-adjusted data for such analyses[6]. However, given the low levels of the endogenous LD, the Applicant's approach is considered acceptable. The reviewer repeated the analyses using baseline-adjusted data and demonstrated similar results.*
- *The ratios of baseline-unadjusted dose-normalized-AUC_{0-24h} and dose-normalized C_{max} following a CVT-301 84 mg dose were approximately 70% and 50%, respectively, of those following SINEMET® 25-100 dose. Similarly, the ratios of baseline-adjusted dose-normalized-*

[6] https://www.fda.gov/downloads/Drugs/GuidanceComplianceRegulatoryInformation/Guidances/UCM389370.pdf

AUC_{0-24h} and dose-normalized C_{max} following a CVT-301 84 mg dose were approximately 67% and 51%, respectively, of those following SINEMET® 25-100 dose. Therefore, the bioavailability of CVT-301 84 mg is ~70% relative to SINEMET® 25-100 mg oral tablets.

- *The ratios of baseline-unadjusted dose-normalized-AUC_{0-24h} and dose-normalized C_{max} following a CVT-301 84 mg dose were approximately 97% and 95%, respectively, of those following a CVT-301 60 mg dose. The 90% CIs of the ratio of the CVT-301 84 mg dose compared to the CVT-301 60 mg dose of all parameters were contained within the 80% to 125% interval. This indicated dose proportionality between the CVT-301 60 mg and 84 mg doses.*

4.3 Effect of Gender on the Pharmacokinetics of INBRIJA™

The applicant performed a post-hoc analysis with data from subjects enrolled in study CVT-301-010 to assess the effect of gender on the PK of CVT-301. There were 13 males and 11 females. A statistically significant difference in LD's exposure parameters between males and females were demonstrated ($p<0.05$). However, these differences are accounted for by body weight differences. The dose normalized, baseline-unadjusted C_{max} and AUC_{0-24} for females and males are presented in Table 5.

Table 5: Summary of Pharmacokinetic Parameters for Levodopa, by Gender

PK Parameter (Unit)	SINEMET 25-100[a] Males	SINEMET 25-100[a] Females	CVT-301 84 mg[a] Males	CVT-301 84 mg[a] Females
Body Weight Unadjusted				
C_{max} (ng/mL)	1480.8 (36.05)	1859.1 (32.10)	602.5 (26.10)	857.1 (31.96)
AUC_{0-24h} (h•ng/mL)	2555.9 (15.88)	3848.4 (23.97)	1484.2 (30.98)	2208.5 (14.42)
DN-C_{max} (ng/mL/mg)	14.8 (36.05)	18.6 (32.10)	7.8 (25.94)	10.2 (31.96)
DN- AUC_{0-24h} (h•ng/mL/mg)	25.6 (15.88)	38.5 (23.97)	17.7 (30.98)	26.3 (14.42)
Body Weight Adjusted				
C_{max} ((ng/mL)•kg)	120851 (33.36)	117000 (40.65)	49174 (31.22)	53938 (23.31)
AUC_{0-24h} ((h•ng/mL/mg) •kg)	208583 (13.01)	242190 (30.09)	120801 (33.01)	138983 (15.51)
DN- C_{max} ((ng/mL/mg) •kg)	1208.5(33.36)	1170.0 (40.65)	585.4 (31.22)	642.1 (23.31)
DN- AUC_{0-24h} ((h•ng/mL/mg) •kg)	2085.8 (13.01)	2421.9 (30.09)	1438.1 (33.01)	1654.6 (15.51)

[a] Geometric Mean (CV%)
Source: 2.7.2 Summary of Clinical Pharmacology Studies, Table 24.

Reviewer Comment:

- *Similar results were obtained when the analysis was repeated using the baseline-adjusted C_{max} and AUC_{0-24h}. For example, after a dose of SINEMET® 25-100 the baseline-adjusted C_{max} and AUC_{0-24h} were 19% and 50.6% higher, respectively, compared to males. After a CVT-301 84 mg dose, females had a 39% higher baseline-adjusted C_{max} and a 48.8% higher baseline-adjusted AUC_{0-24h} when compared to males.*
- *Similar findings are documented with other levodopa products[7,8]. No dosing adjustment is necessary based on gender differences.*

4.4 Effect of Smoking on the Pharmacokinetics of INBRIJA™

Study Report: CVT-301-007

Title of the Study: A Phase 1 Study of the Safety and Levodopa Pharmacokinetics Following Single Dose Administration of CVT-301 (Levodopa Inhalation Powder) in Smoking and Non-Smoking Adults

Objectives:

The primary objective of the study was to characterize the PK of LD following a single dose of CVT-301 84 mg in smoking and non-smoking adults.

Study Design:

This study was an open-label parallel group study in 2 groups of subjects: smoking and non-smoking adults. The CVT-301 formulation that was used in this study is the to-be marketed formulation.

Subjects were treated with 50 mg carbidopa that was administered every 8 hours from the morning of Day -1 until Day 1. On Day 1, subjects received a single final dose of carbidopa, and approximately 1 hour later received a single dose of CVT-301 84 mg.

Smokers continued to smoke throughout the study at a level consistent with the typical amounts per day for the individual during the 12 months prior to study participation. However, an interval of 1 hour was imposed between smoking and CVT-301 administration.

Population:

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[7] https://www.accessdata.fda.gov/drugsatfda_docs/label/2010/021485s20lbl.pdf
[8] https://www.accessdata.fda.gov/drugsatfda_docs/label/2015/203312s000lbl.pdf

Healthy men or women aged between 25 and 65 years. Smokers had to have smoked ≥10 cigarettes/day for at least 12 months. Twenty-eight smokers and 35 non-smokers were enrolled and data for 25 smokers and 31 non-smokers were analyzed.

PK and Blood Sampling:

Blood samples (a total of 14) were collected over 24 hours for the analysis of plasma concentrations of levodopa.

Statistical Methods:

Analysis of variance (ANOVA) was used for the analysis of the log-transformed levodopa PK parameters ($AUC_{0\text{-}24h}$, $AUC_{0\text{-}\infty}$, and C_{max}); subject group was a fixed effect in the model. LSM for each subject group, and geometric mean ratios of LSM were calculated for the log-transformed PK parameters together with their associated 90% confidence intervals (CIs) using the residual variance derived from the ANOVA model. The ratio of geometric LSM of smokers and non-smokers was compared and its 90% CIs were calculated.

Results:

Following inhaled doses of CVT-301, absorption of LD was rapid in both smokers and non-smokers. Median T_{max} was 30, and 45 minutes post-dose in non-smokers and smokers, respectively. Geometric mean $t_{1/2}$ for levodopa was also similar in both groups, at approximately 1.6 hours. The mean C_{max} and $AUC_{0\text{-}24h}$ were slightly higher in smokers compared to non-smokers (Table 6).

Table 6: Statistical analyses of the effects of smoking on the PK parameters of levodopa are presented below

PK Parameter (Unit)	Non-smokers[a]	Smokers[a]	Smokers vs. Non-smokers		
			Ratio	90% CI	CV%
$AUC_{0\text{-}24h}$ (ng•hr/mL)	1368 (31)	1523 (25)	111.3	91.7, 135.2	45.2
$AUC_{0\text{-}\infty}$ (ng·hr/mL)	1350 (30)	1497 (24)	110.9	91.0, 135.1	45.2
C_{max} (ng/mL)	684 (31)	765 (25)	111.9	93.4, 134.1	41.9

[a] LSM

Source: CVT-301-007 CSR, Table 9.

Reviewer Comments:

The overall study design is acceptable. The results indicate that no dose adjustment is necessary in smokers.

4.5 Effect of Asthma on the Pharmacokinetics of INBRIJA™

The effect of asthma on the PK of INBRIJA™ was investigated as a secondary objective in the Phase 1 study CVT-301-008. The primary objective of this study was to investigate the pulmonary safety in adults with asthma after multiple inhaled doses of CVT-301 84 mg.

Study CVT-301-008 was a double-blind, randomized, placebo-controlled, 2-period, crossover study in asthmatic adults (N=25). The study was conducted using the to-be-marketed formulation. Subjects received 3 doses of CVT-301 84 mg and 3 doses of placebo in 2 dosing periods separated by 1 day. The 3 doses were administered every 4 hours. Carbidopa was administered orally at 50 mg approximately every 8 hours throughout the study from Day -2 to Day 4.

Rich PK sampling was collected only after the first dose and only sparse PK blood sampling was collected following the second and third doses. Therefore, the values of $AUC_{0\text{-}24h}$ were omitted in Table 7. Summary of PK parameters adjusted and unadjusted for the pre-dose LD concentration are reported in the same Table.

Table 7: Summary of Pharmacokinetic Parameters for Levodopa in Adults with Asthma

PK Parameter (Unit)	Geometric Mean (CV%) (N=24)	
	Baseline-unadjusted	**Baseline-adjusted**
$AUC_{0\text{-}4h}$ (ng•h/mL)[a]	1320 (31.3%)	1270 (31.7%)
C_{max} (ng/mL)[a]	655 (38.0%)	642 (39.0%)
T_{max} (min)[a,b]	47 (19 - 118)	47 (19 - 118)
$t_{1/2}$ (h)[a]	1.37 (19.3%)[c]	1.31 (22.3%)
CL/F (L/h)	54.9 (27.9%)[c]	57.2 (28.1%)

Note: Geometric mean (CV%) data are presented except for T_{max}.
[a] Values reported are relative to the first dose.
[b] Median (range).
[c] N=17.
Source: CVT-301-008 CSR, Table 7.

Reviewer Comments:

Overall, the PK of levodopa was similar between subjects in previous studies and asthmatic subjects in this study. Because of the risk of bronchospasm in patients with asthma, the applicant is proposing not to recommend INBRIJA™ in this special population. Please refer to the clinical review for more details.

This is a representation of an electronic record that was signed electronically. Following this are manifestations of any and all electronic signatures for this electronic record.

/s/

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12/07/2018

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12/07/2018

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